

Gen Ed Reflection

Engineering at Iowa State has taught me high-level mathematics, operating system design, and computer architecture. While those topics are fundamental to my major, I believe my general education classes: Econ 101, Psych 280, Arch 321, and U ST 105/106 enabled me to think more broadly, communicate efficiently, and understand the impact my decisions have on the world.

In Econ 101, Principles of Microeconomics, concepts such as resource allocation and opportunity cost have become more apparent as I complete higher-level engineering classes. When implementing a CPU in VHDL or programming a microcontroller for a Roomba, I've had to evaluate the tradeoffs of programming decisions and designs. Econ helped me understand the advantages of engineering teams, enabling people to work individually at times and as a group when a project demands it. When considering design choices, economics relates directly to supply and demand by understanding how changes in trends or consumer demand impact the scope of a project. Econ has enabled me to approach engineering situations from a different perspective.

Psych 280, Social Psychology, enabled me to think about how the environment, perception, and society around them shape human behavior. As an engineer, understanding human behavior is crucial for design choices. Initially, I had a misconception that engineers only design for machines or systems. Still, my coursework has taught me that people are at the center of all design decisions. Understanding how people make judgments based on experience, how biases impact workflows, and how social influence alters behavior significantly shapes how everything is designed. The study of human behavior has helped me become a more empathetic engineer by teaching me the importance of a "human first" design approach.

Architecture 321, the History of American Cities, was personally my favorite gen ed class. My professor, Andrew Gleeson, made the class an exciting one with his approach to teaching and passion for architecture. In this class, I learned about the reasons cities remain rooted so close to rivers, how supply chain needs shaped a city's infrastructure, and the discrimination that minority communities constantly faced when seeking housing. Gleeson strongly argued that urban renewal projects produced promising infrastructure reforms but also displaced vulnerable communities. His teachings broadened my awareness about the ethical considerations I need to make as an engineer. As I progress in my career, I will become a professional who acknowledges and weighs the outcomes of my decisions to ensure they align with my personal virtues.

U ST 105/106, George Washington Carver Seminar, was the most formative general education class I had the opportunity to take. This scholarship program enabled me to connect with peers and mentors through group activities, crucial conversations, and research about classism in education. This program challenged me to think critically about my role as an engineering student and as a member of my community. Learning about George Washington Carver's legacy, his impact on agricultural productivity, and his relentless drive to constantly learn was an unforgettable experience. The discussions and opportunities this program provided me opened my eyes to what it means to be an engineer from an underrepresented background and to the valuable perspective I bring to engineering decisions.

Across these four courses, I encountered new ideas, perspectives, and ethical considerations that I will face as an engineer. The design choices and problems I face will never be purely technical; they will always involve some human interaction, and I must be mindful of it. I believe general education courses are integral to an engineering program, broadening students' perspectives beyond the technical world. Combined with technical coursework, an engineer becomes more well-rounded and develops ethical awareness, empathy, and a lifelong commitment to addressing the engineering challenges society faces.